Sebastián Ignacio Allende Cuello

Odoo/ERP Engineer & Applied AI

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PROFESSIONAL SUMMARY

Computer & Telecommunications Engineer with hands-on experience in Odoo 14–19 and automation with Artificial Intelligence. Migrated a production ERP for 4 companies across 3 countries to Odoo 19, with invoices, orders and stock verified against the previous version, and manage multi-country e-invoicing (SII, DIAN, SUNAT). Work with automated tests, staged releases and production monitoring.

WORK EXPERIENCE

IT Manager & ERP Administrator December 2025 – Present

TOYNG, Importadora y Comercializadora Toyng Ltda. | Santiago, Chile (hybrid)

- Migrated the production ERP from Odoo 14 to Odoo 19 (June 2026): 4 companies across 3 countries, about 228 installed modules, and 41 custom modules consolidated into 18.
- Validated the cutover against Odoo 14: identical invoices, orders and deliveries, and millions of stock units reconciled.
- Found a silent upgrade bug and repaired over 500,000 invoices; closed or logged 16 issues in 3 days after go-live.
- Manage multi-country e-invoicing (SII, DIAN, SUNAT), with electronic ledgers and batch electronic waybills in Peru.
- Maintain retail consignment (Falabella, Ripley, Éxito), the automated inter-company flow and integrations with Mercado Libre and Yuju.
- Built automated monitoring: daily health check with over 40 detectors and urgent alerts every 30 minutes.
- Backed critical flows with about 180 test files and PostgreSQL invariants that block inconsistent data.
- Organized development and releases: a development branch, staging before production, and production deploys outside business hours in all 3 countries.

Technologies: Python, Odoo, PostgreSQL, JavaScript, Git, Odoo.sh.

Security and Identity Management August 2025 – December 2025

Banco BICE | Chile

- Managed access and the identity lifecycle across Active Directory, Salesforce and critical banking systems.
- Took part in the mass migration of accounts for a bank merger and in proofs of concept for fixed-term deposits.

Independent Digital Solutions Consultant November 2024 – Present

Clients: Gelymar, Farmacia Doctor Simi | Chile

- Gelymar: rebuilt the logistics system from scratch (Docker, SAP). Doctor Simi: optimized the infrastructure.

Business Transformation Consultant, CX Analyst | Imppulsor, Ireland (remote) 2024

Professional Internship 2023

Gtd Manquehue S.A. | Santiago, Chile

- Document reading and summarization flow with Power Apps and OpenAI; BPMN model for IT license deprovisioning.

Teaching Assistant, Advanced Programming and Databases | Universidad Finis Terrae 2022

SKILLS

ERP and Security: Odoo 14–19, Odoo.sh, module development, migrations, e-invoicing (SII, DIAN, SUNAT), Mercado Libre, Yuju, SAP S/4HANA, Salesforce, Active Directory, identity and access management (IAM), server hardening.

Languages and Databases: Python, TypeScript, JavaScript, SQL, Rust, Java, PHP, C, R; PostgreSQL, Prisma, Supabase, SQLite.

Testing and DevOps: pytest, Vitest, Playwright, pgTAP, Git, GitHub Actions, continuous integration (CI), Docker, Linux, Cloudflare Workers, Azure, AWS, HPC (NLHPC).

Artificial Intelligence: Generative AI, large language models (LLMs), Anthropic API (Claude), Vercel AI SDK, Gemini, OpenAI, Ollama, tool calling, structured outputs, vision and PDF (multimodal), guardrails, prompt engineering, RAG, embeddings, hybrid search, pgvector, LLM evaluation (recall@k, MRR).

AI Agents: Agent orchestration, Model Context Protocol (MCP), voice interfaces.

Backend, Frontend and Mobile: FastAPI, Axum, Fastify, Hono, REST APIs, OpenAPI, webhooks, Next.js, React, React Native, Expo, SwiftUI.

PROJECTS

Odoo 19 Migration Field Notes: failures from a real migration github.com/Ssebv/odoo-19-migration-field-notes

- 13 failures from a real Odoo 14 to 19 upgrade that raise no error, each written up as symptom, cause and fix.

Odoo RAG: assistant over technical docs, with evaluation Code: github.com/Ssebv/odoo-rag

- RAG with hybrid search (vector and keyword), query translation and cited answers, using local models.
- Evaluated on 45 hand-written questions: recall@5 of 82% vs. 67% vector-only; in Spanish, from 20% to 90%.
- Answers: 90% with expected terms, 80% with valid citations, and 7 of 8 out-of-scope questions refused.

Technologies: Python, PostgreSQL, pgvector, Ollama, Docker, pytest, GitHub Actions.

Citofonía: the building intercom on your phone Case study: ssebv.github.io/projects/citofonia

- React Native app for residents and the concierge, plus a web panel: calls that ring on several phones and go to the first to answer, QR visits and parcels (in development).
- Notifications carry no names and the audit log has no personal data; 105 migrations, 88 pgTAP files, E2E tests in CI.

Technologies: TypeScript, React Native, Expo, Hono, Supabase, LiveKit, Cloudflare Workers, pgTAP, Vitest.

ModoMax: online store with AI assistants Site: modomax.ressc.vip

- Shopping assistant with streaming, tool-based search and usage limits; chose tool calling over RAG for catalog size.
- WhatsApp support that discards a reply if another message arrives while the model is processing.
- Admin panel with chat, product descriptions and business insights validated with Zod schemas.

Technologies: TypeScript, Next.js, React, Prisma, PostgreSQL, Vercel AI SDK, Anthropic API, Vitest, Playwright.

FlagForge: feature flags and A/B experiments in Rust Code: github.com/Ssebv/flagforge

- Multi-tenant: segment-targeted rollouts, deterministic bucketing and an audit trail; in-memory evaluation under 1 µs.
- 245 tests in four layers and CI; API documented with OpenAPI and a dashboard in Rust compiled to WebAssembly.

Technologies: Rust, Axum, PostgreSQL (sqlx), Leptos (WebAssembly), OpenAPI, Docker, GitHub Actions.

Construye Hoy: AI site control for a construction firm, in pilot Case study: ssebv.github.io/projects/construye-hoy

- Validated extraction from images and PDFs (invoices, quantity take-offs), with cost computed per model and call.
- Swappable LLM provider and tests using recorded responses, with no network; 101 test files and CI.

Technologies: TypeScript, Hono, Cloudflare Workers, Supabase, Expo, React, Anthropic API, Vitest, GitHub Actions.

SAC Next: community alarms, in development Case study: ssebv.github.io/projects/sac-next

- Assistant restricted to its knowledge base, with personal data redaction and emergency detection.
- ~100-route API, one deployment per client, encrypted backups; tested on data for 45 panels and 1,100 homes.

Technologies: TypeScript, Node.js, Fastify, PostgreSQL, MQTT, Expo, React, Docker, Anthropic API, Vitest.

EDUCATION

Computer & Telecommunications Engineer Graduated 2026

Universidad Finis Terrae | Santiago, Chile Code: github.com/Ssebv/TBSimAlgoritmoGenetico_HPC_LOCAL

Chilean degree: Ingeniero Civil en Informática y Telecomunicaciones. Thesis: Computational scalability of parallel evolutionary algorithms, applied to robotic soccer teams.

Studies abroad: English as a Second Language (ESL) 2020 – 2021

Mississauga Secondary Academy | Ontario, Canada

LANGUAGES

Spanish: native | English: upper-intermediate (B2)